

PEFT 参数高效微调



- LLM 微调理论
- BitFit
- Prompt Tuning
- P-Tuning
- Prefix Tuning
- LoRA
- IA3
- PEFT 进阶操作

目录

Content

01

微调、提示词工程、RAG

02

传统大模型训练流程

03

SFT 与 RAG 区别

04

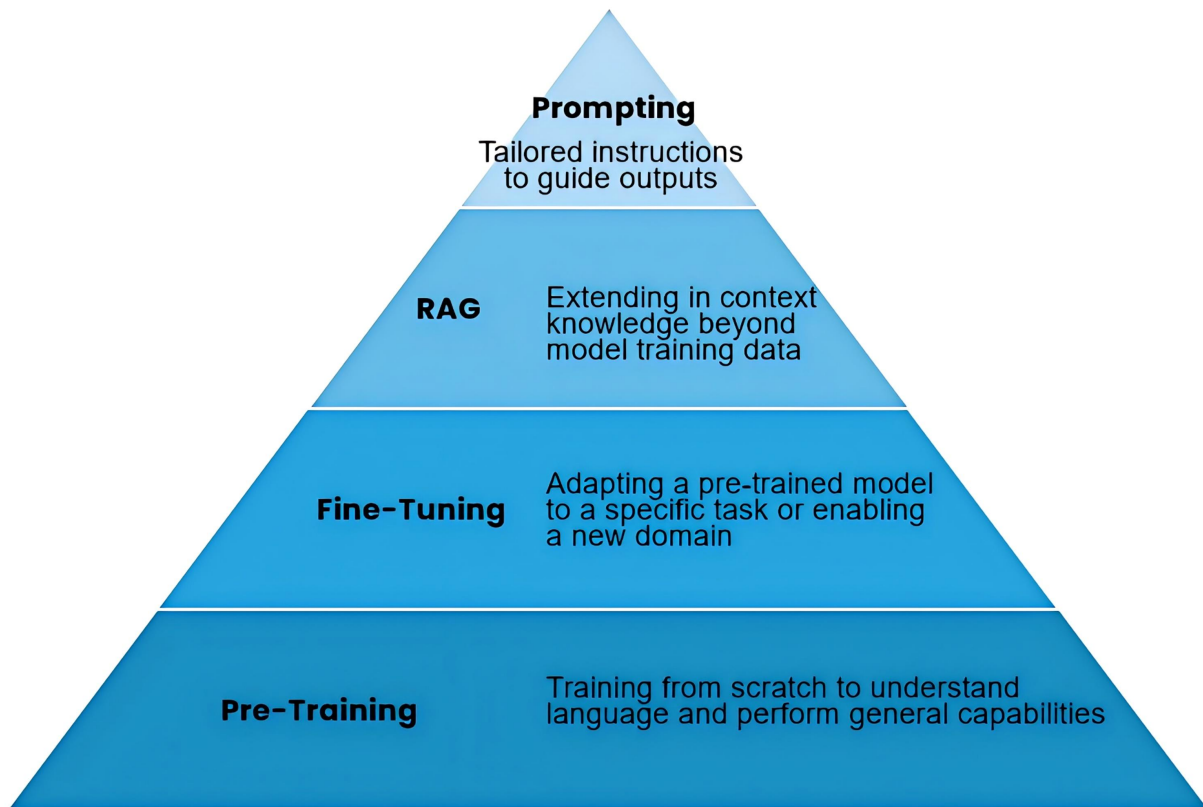
指令数据集构造方式

05

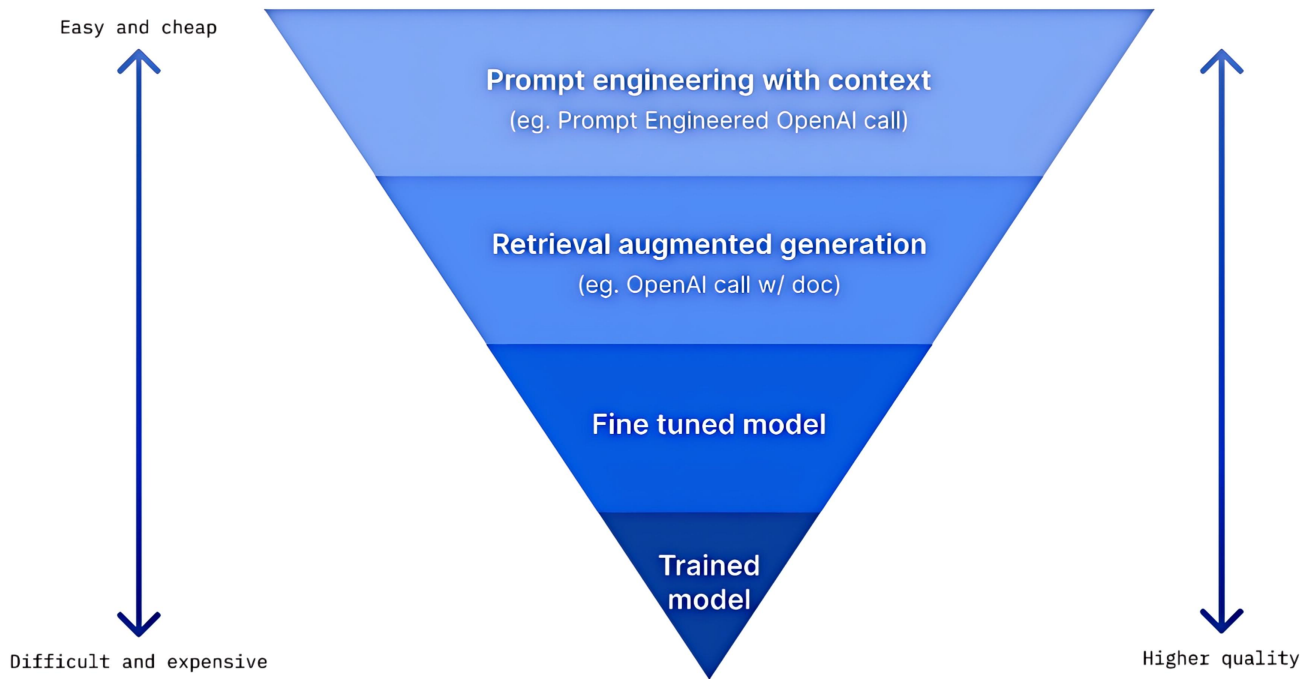
PEFT 与传统微调区别

06

PEFT 的各种方法



GenAI techniques from most complex & compute intensive (bottom) to least (top)



目录

Content

01

微调、提示词工程、RAG

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03

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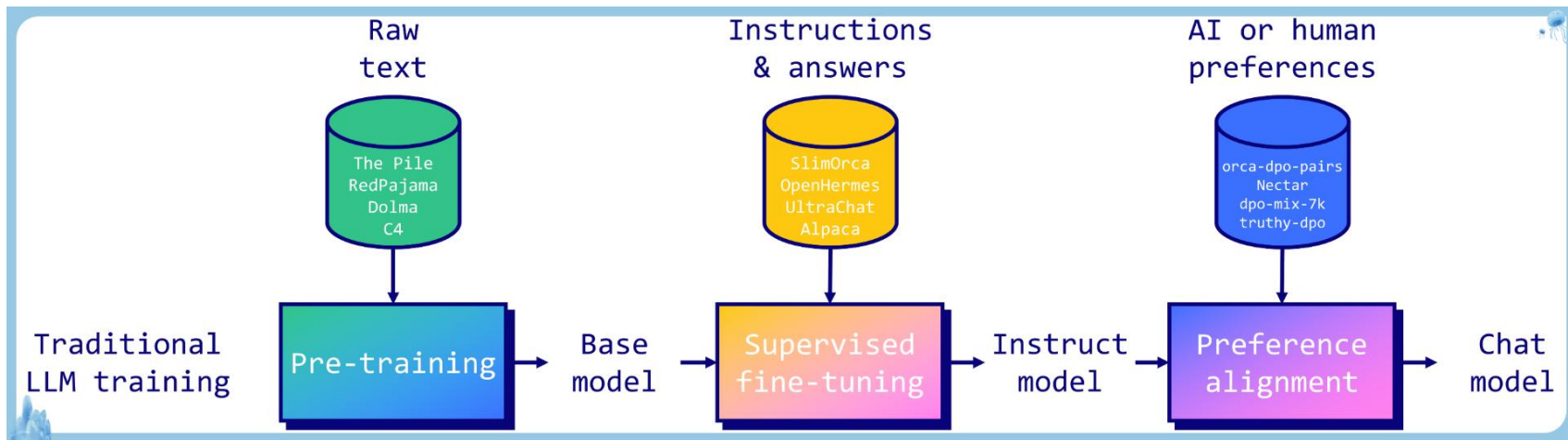
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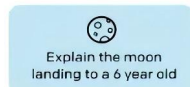
PEFT 的各种方法



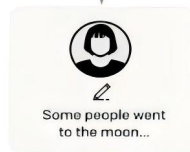
Step 1

Collect demonstration data, and train a supervised policy. **SFT**

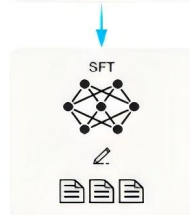
A prompt is sampled from our prompt dataset.



A labeler demonstrates the desired output behavior.



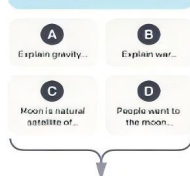
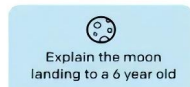
This data is used to fine-tune GPT-3 with supervised learning.



Step 2

Collect comparison data, and train a reward model.

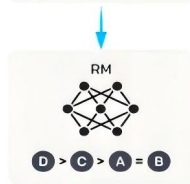
A prompt and several model outputs are sampled.



A labeler ranks the outputs from best to worst.



This data is used to train our reward model.



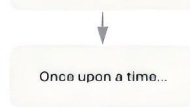
Step 3

Optimize a policy against the reward model using reinforcement learning. **RLHF**

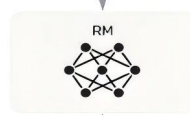
A new prompt is sampled from the dataset.



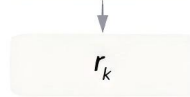
The policy generates an output.



The reward model calculates a reward for the output.



The reward is used to update the policy using PPO.



目录

Content

01

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02

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04

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05

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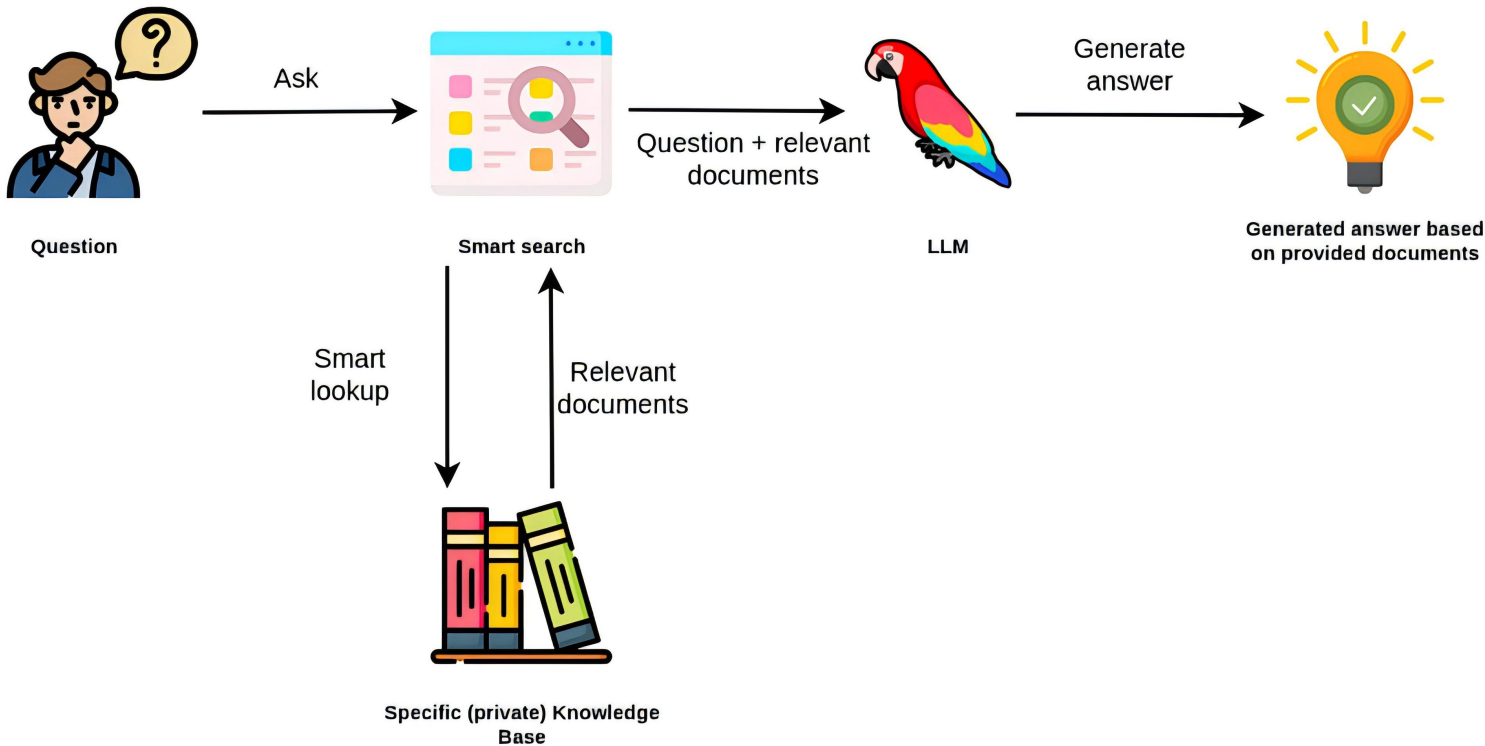
Supervised fine-tuning vs RAG

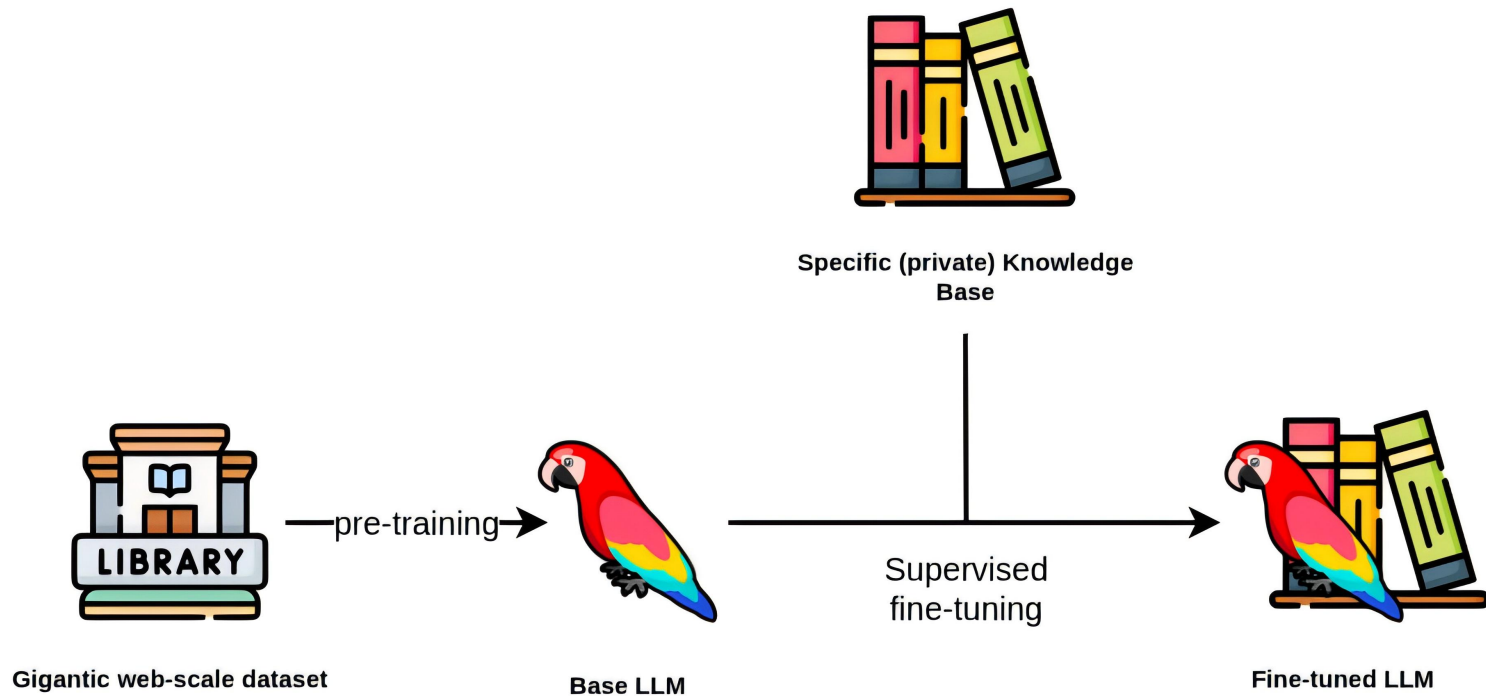
SFT

Helps the LLM to perform well on specific tasks, by providing sample data.

RAG

Helps the LLM to generate more domain-relevant responses, by incorporating additional knowledge sources.





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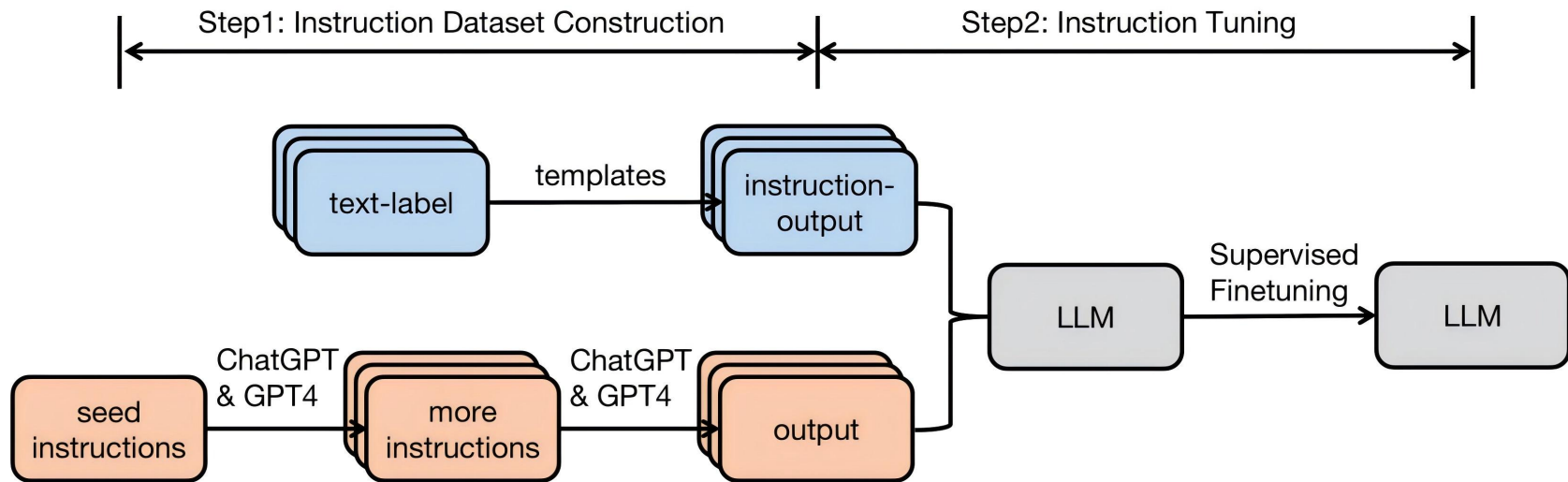
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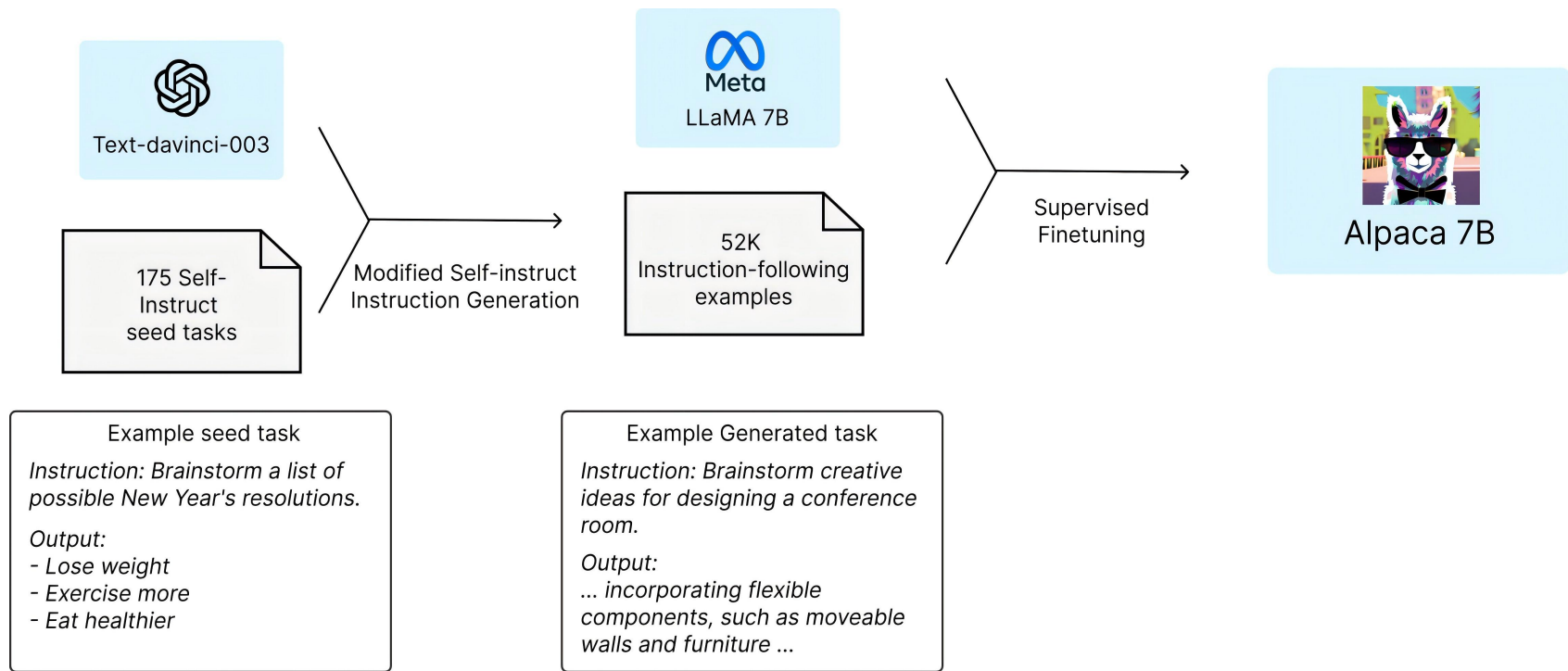
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Content

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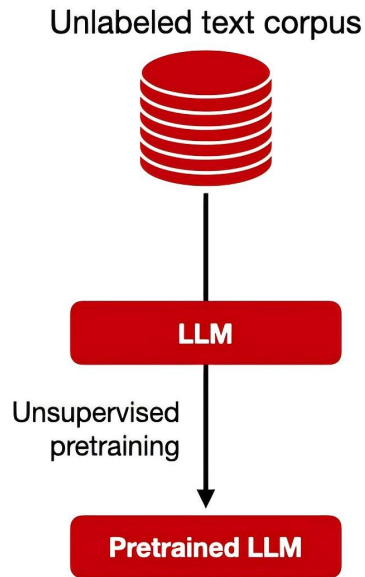
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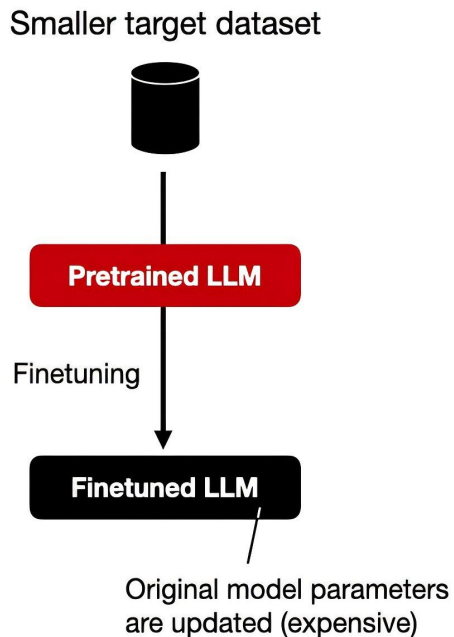
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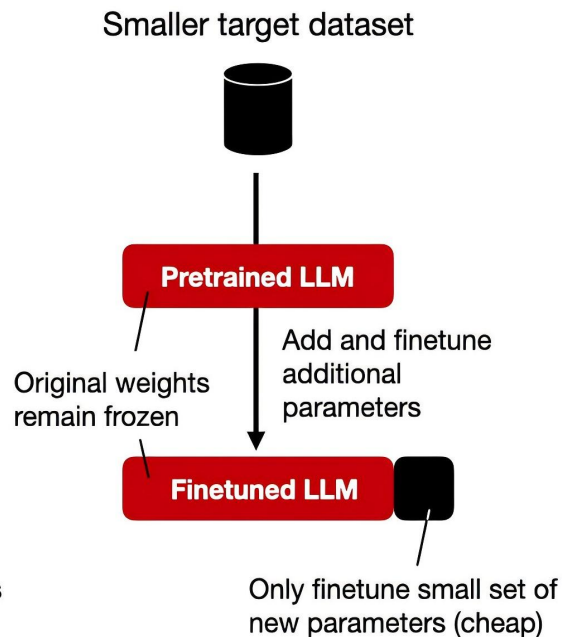
Pretraining



Conventional finetuning



Parameter-efficient finetuning



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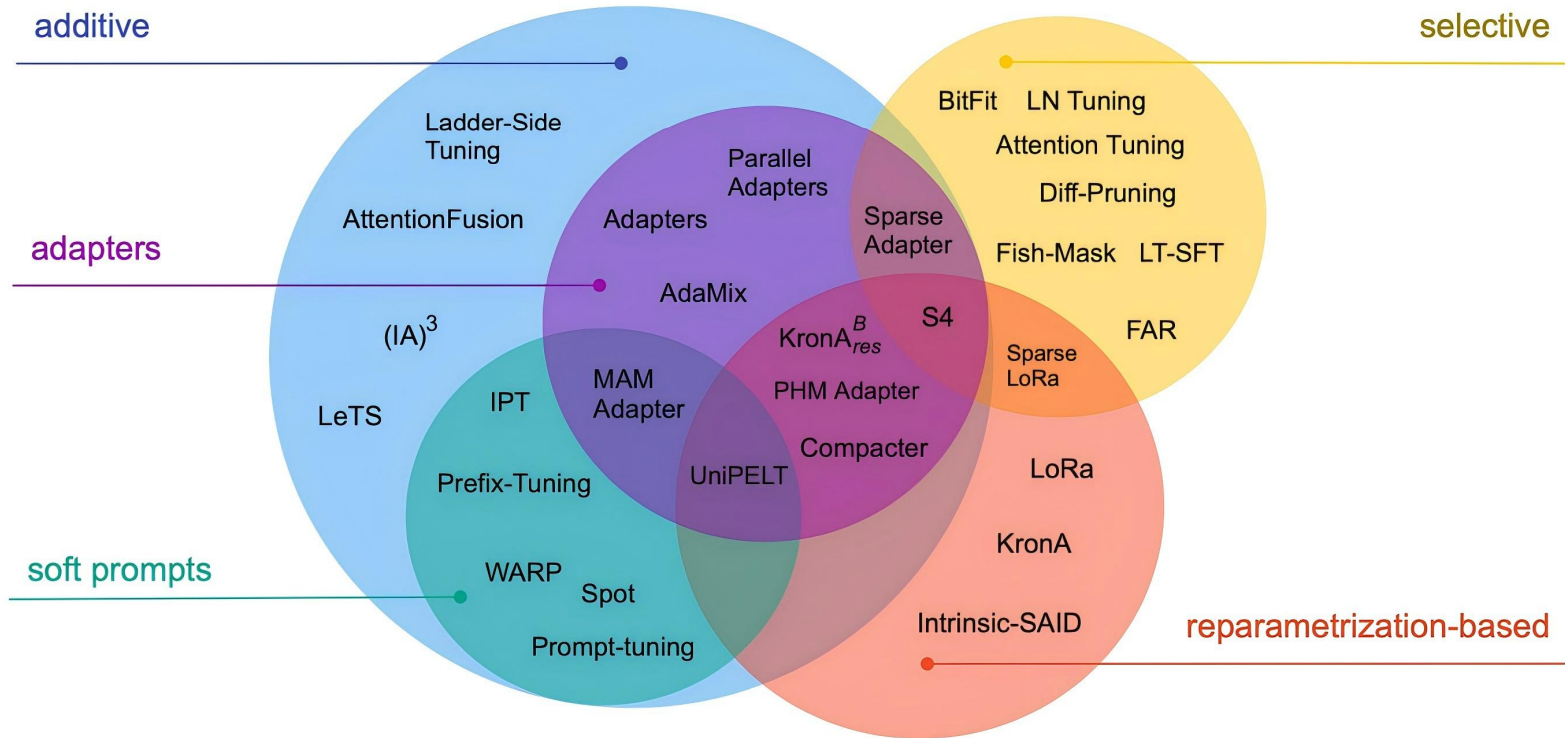
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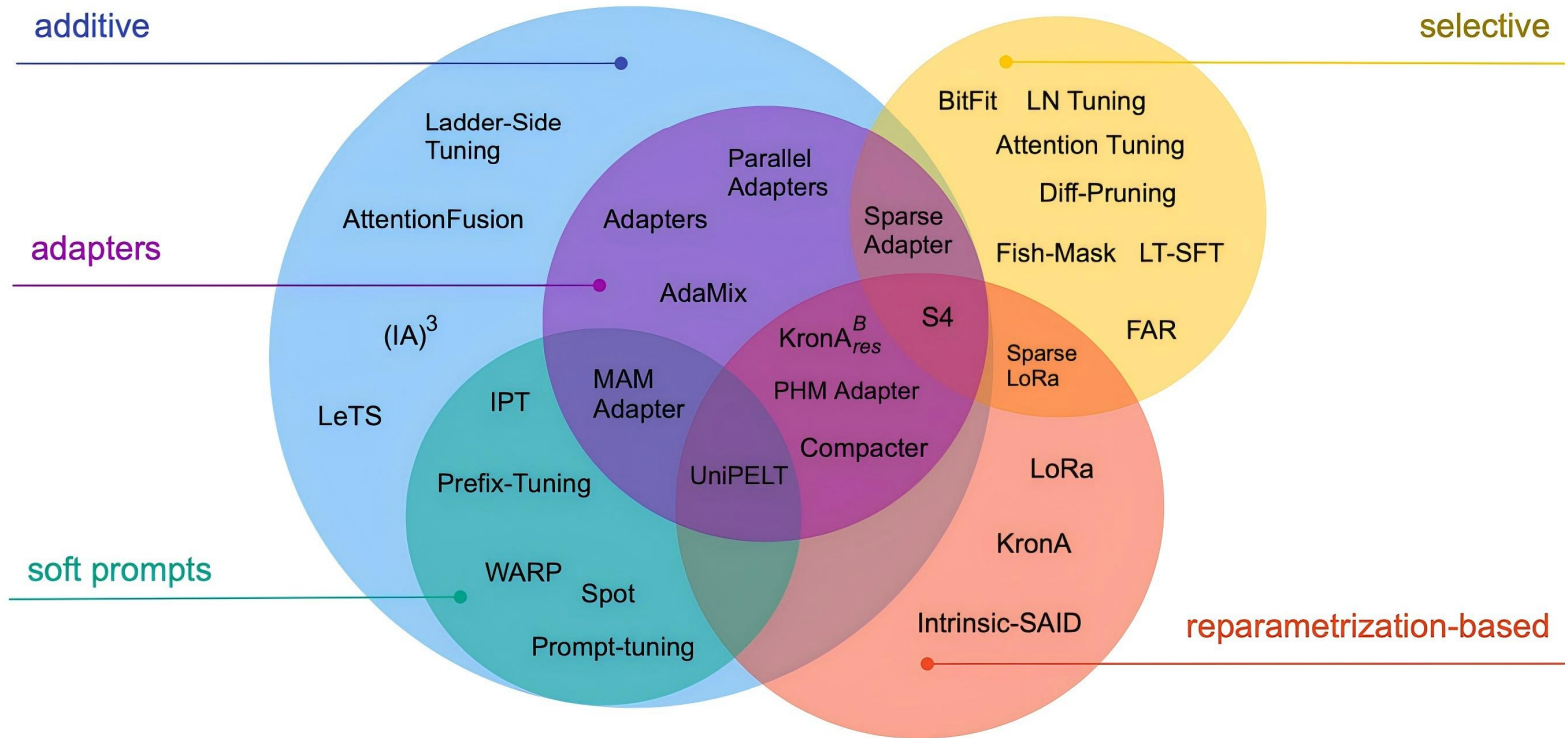
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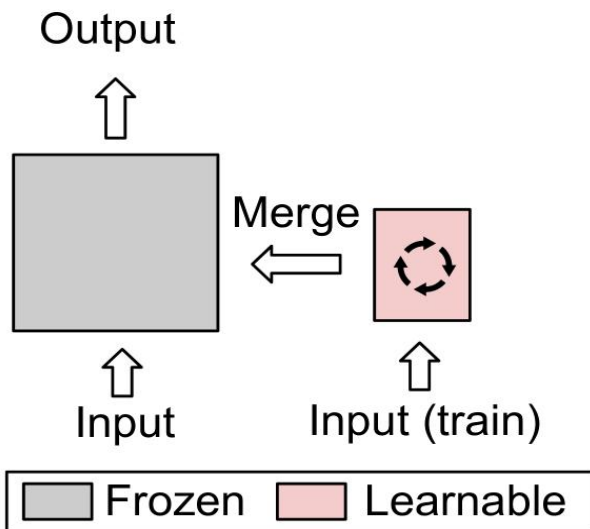
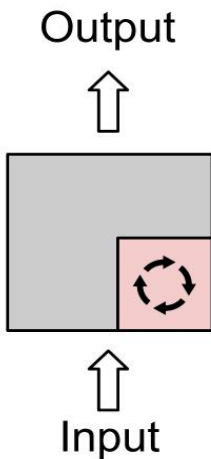
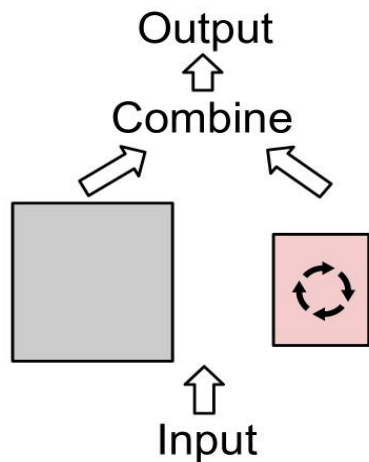
PEFT 参数高效微调

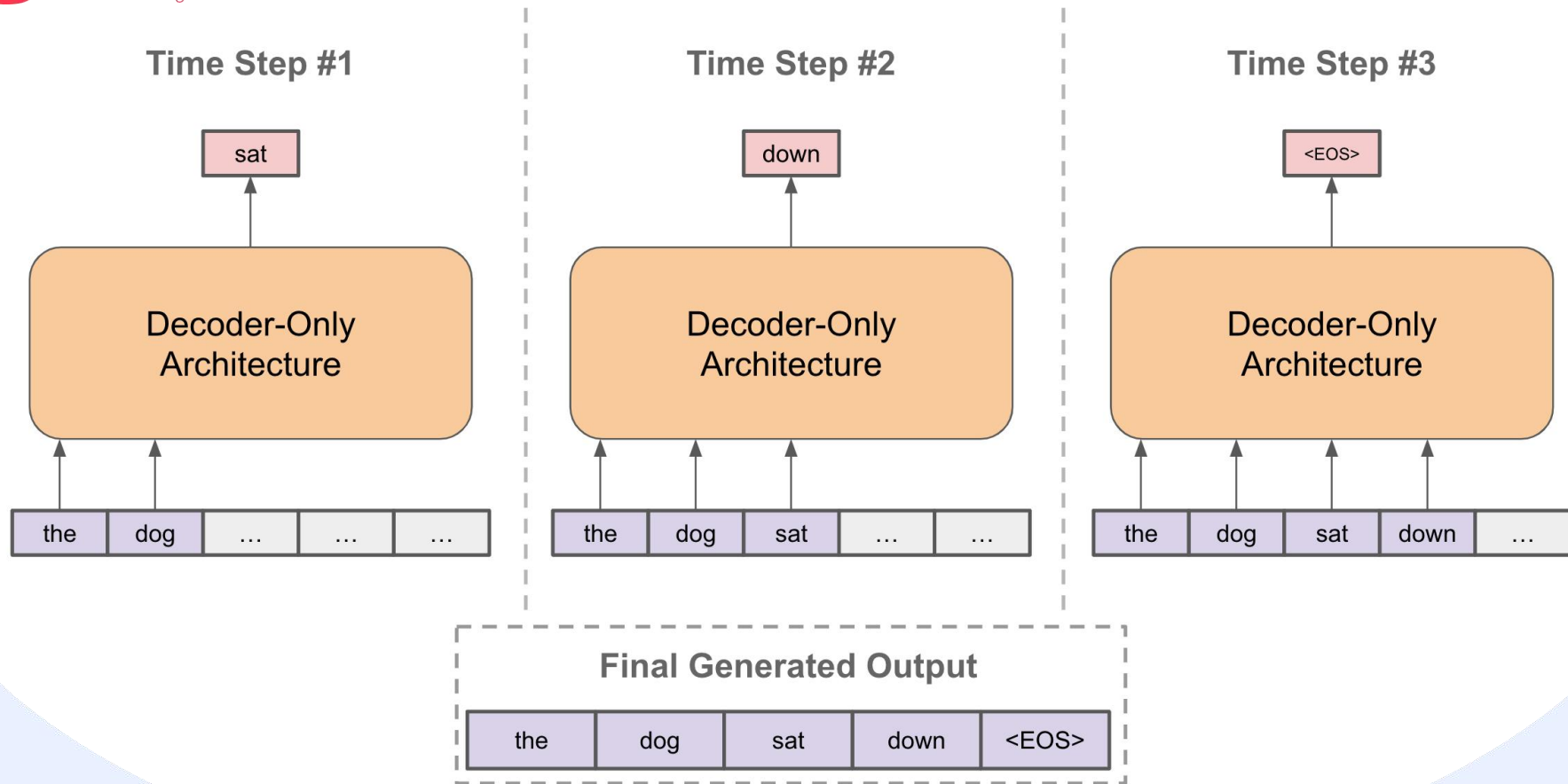


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(a) Additive PEFT (b) Selective PEFT (c) Reparameterization PEFT

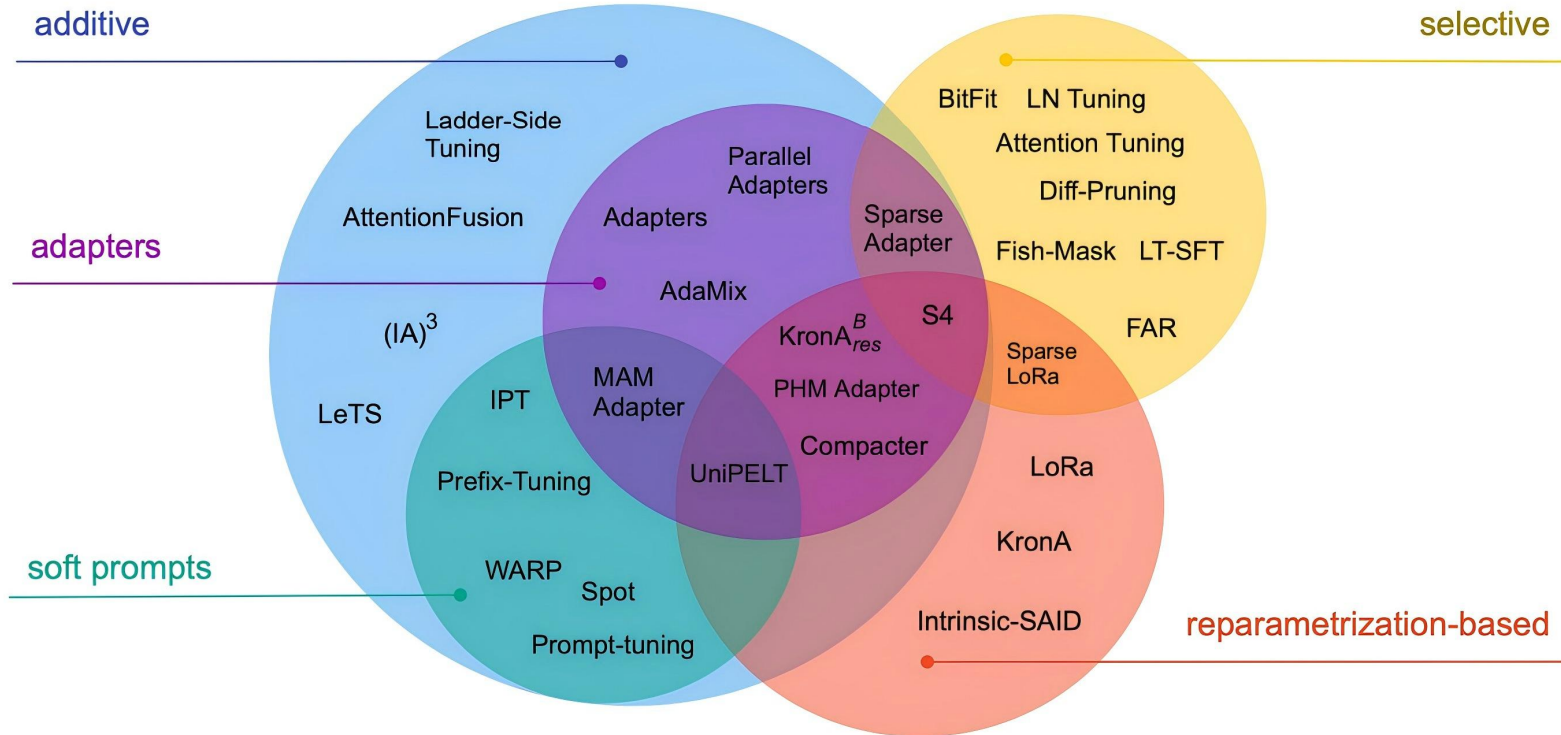




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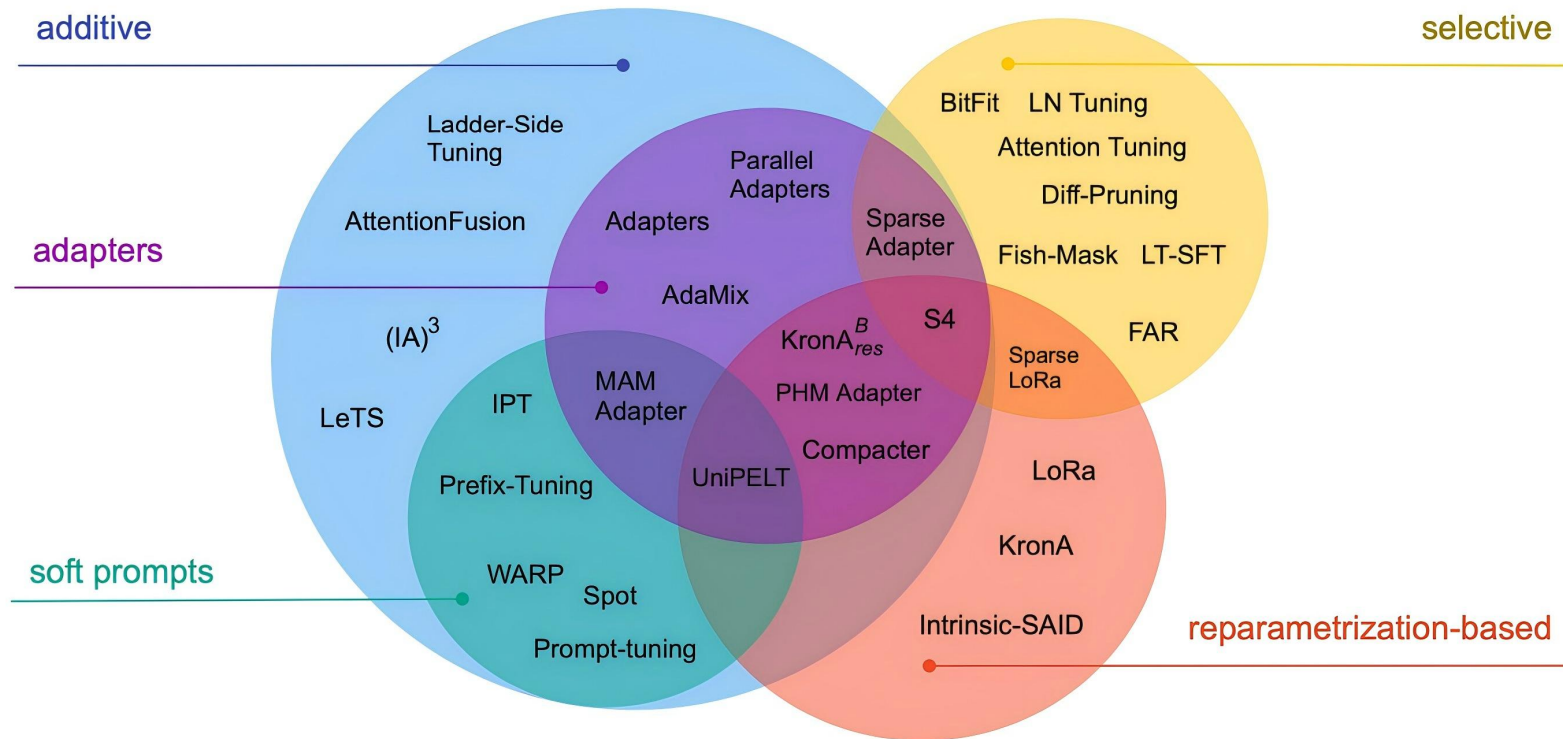


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Additive:

- 在模型中添加新的可调参数层
- 保持基模型参数冻结，仅更新添加的层参数



Soft prompt tuning:

- 拼接可训练参数到输入嵌入层 (input embeddings)

Text Input: In what country is Karlskrona located?

Prompt Engineering

Discrete Manually
Generated Prompt

Text Input

Answer the question:

In what country is Karlskrona located?

Prompt Learning

Automatically Optimized
Virtual Prompt

Tokenized Text Input

t_1 t_2 ... t_n

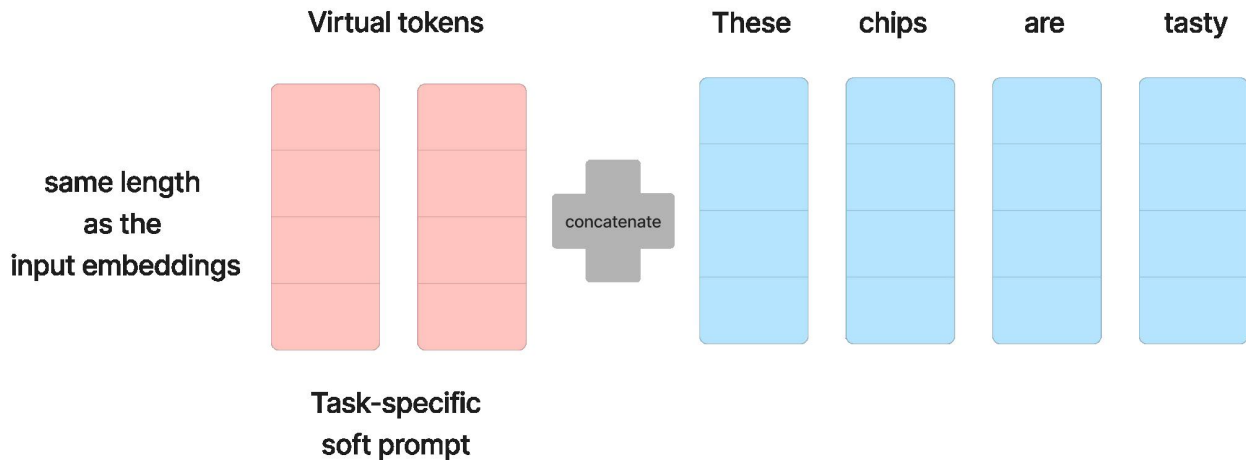
In what country is Karlskrona located ?

Virtual Token

拼接可训练参数到 input embeddings:

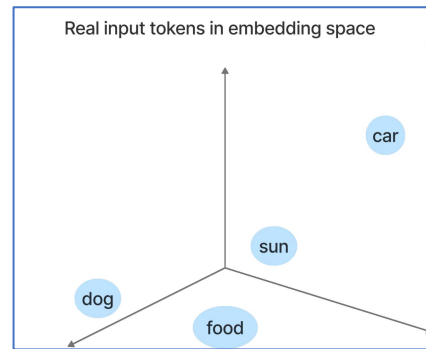
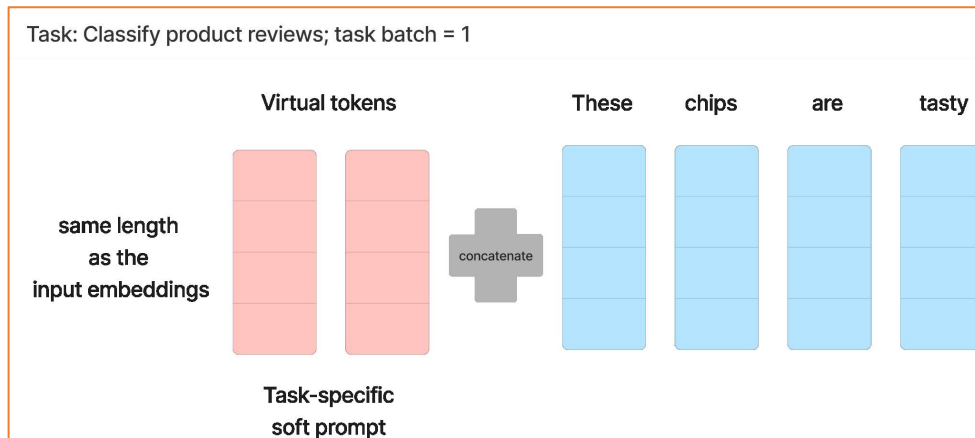
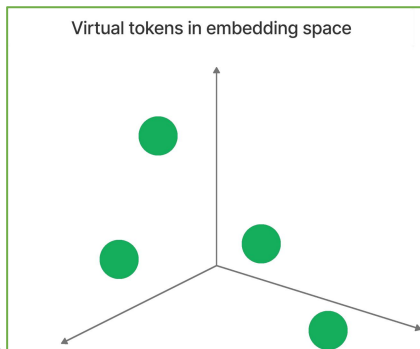
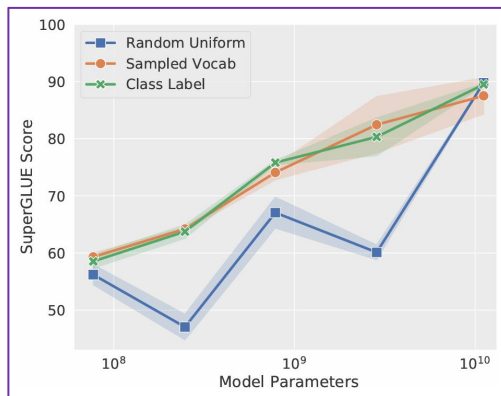
- ✓ 学习一个新的具体任务的 embeddings 序列
- ✓ 我们称其为 prompt tuning 而并非 model tuning 因为我们仅更新 prompt weights

Task: Classify product reviews; task batch = 1



随机初始化 embedding 向量:

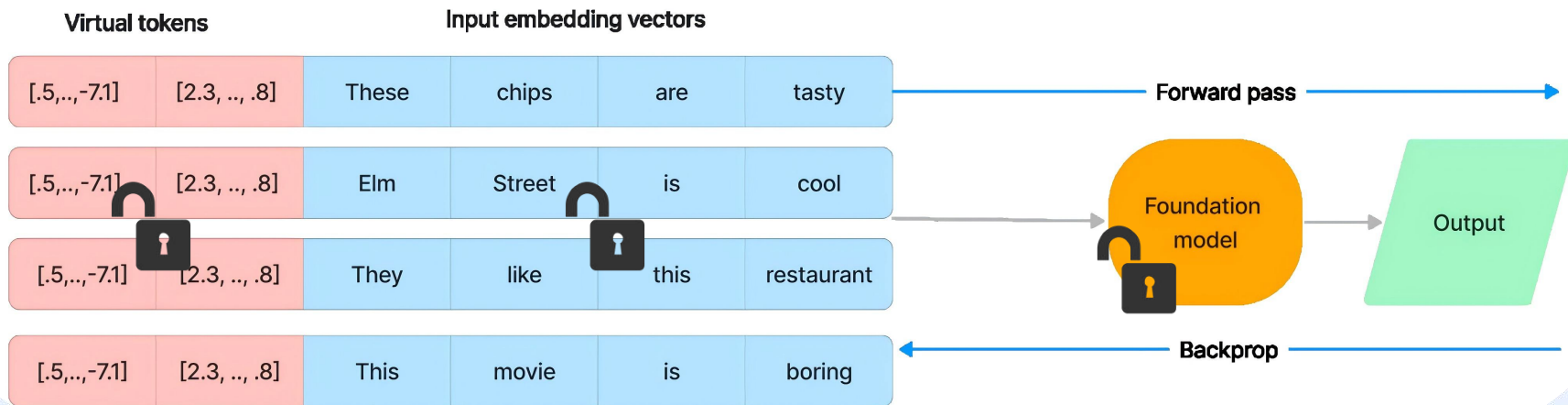
- ✓ 我们也可以初始化为一些具体提示词的向量
- ✓ 但是随机初始化也可以和具体提示词向量几乎一样好
(<https://arxiv.org/pdf/2104.08691>)



全量微调

Backprop: update **all** weights based on loss

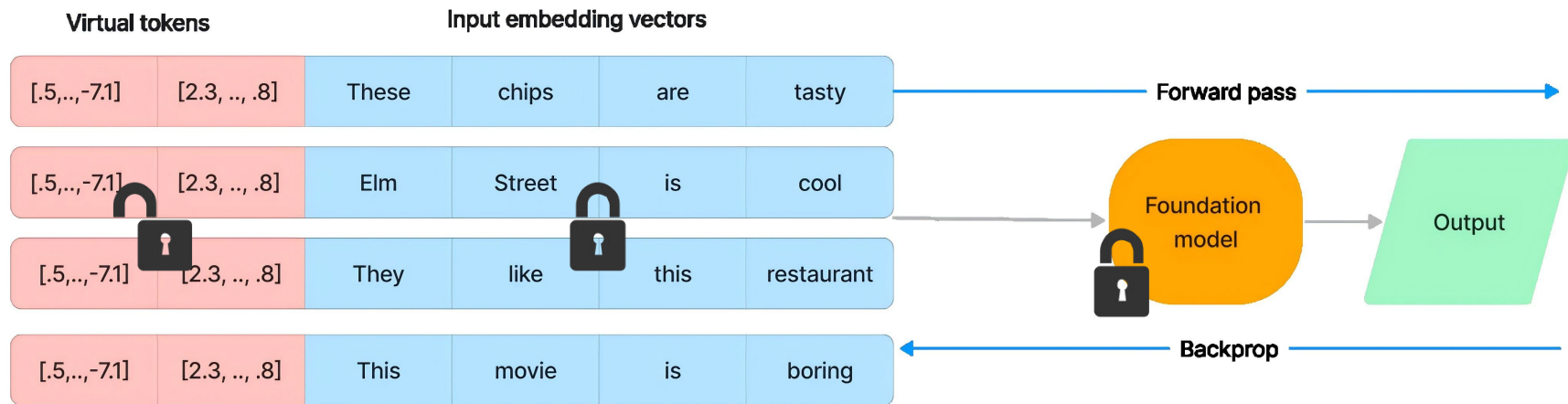
Task: Classify sentiment; task batch = 4



Prompt Tuning

Backprop: update *only prompt* weights based on loss

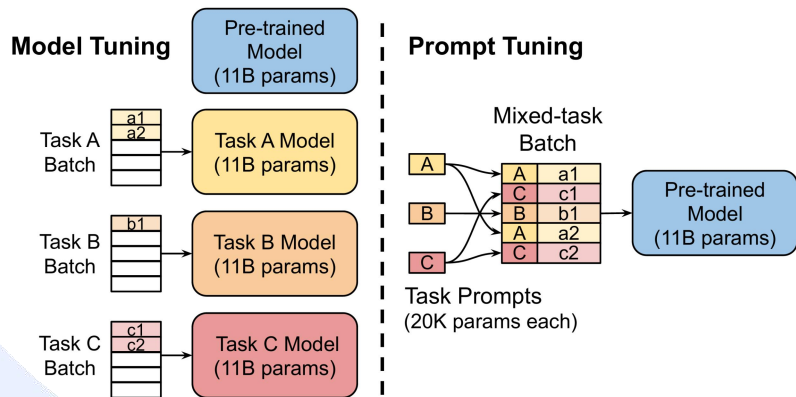
Task: Classify sentiment; task batch = 4



模型自动学习 prompt 最优表达

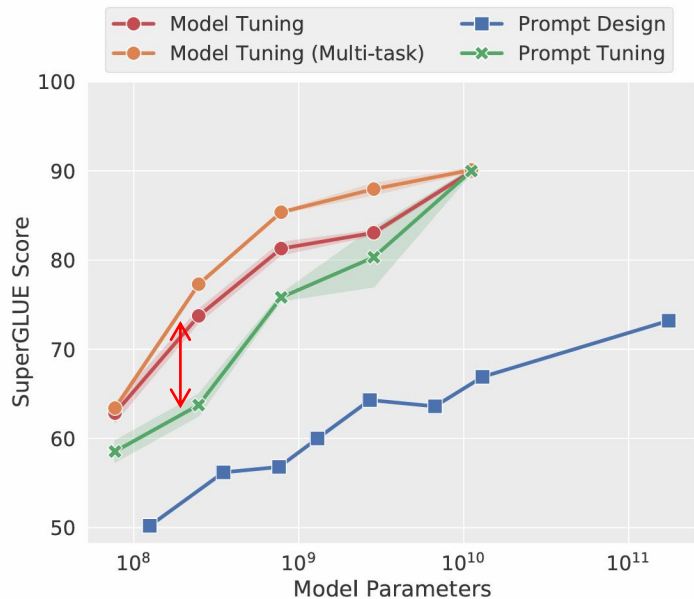
Efficient for multi-task serving

- ✓ 可以切换任务提示词
- ✓ 每个任务对应一个 prompt, 而不是对应一个模型
- ✓ 仅需要 serve 一个冻结的模型 for multi-task serving
- ✓ 对于各种不同任务应用不同的提示词, 同一个任务的不同输出应用同一个提示词
- ✓ 一个 serving request 可以是一个大的混合的任务批次

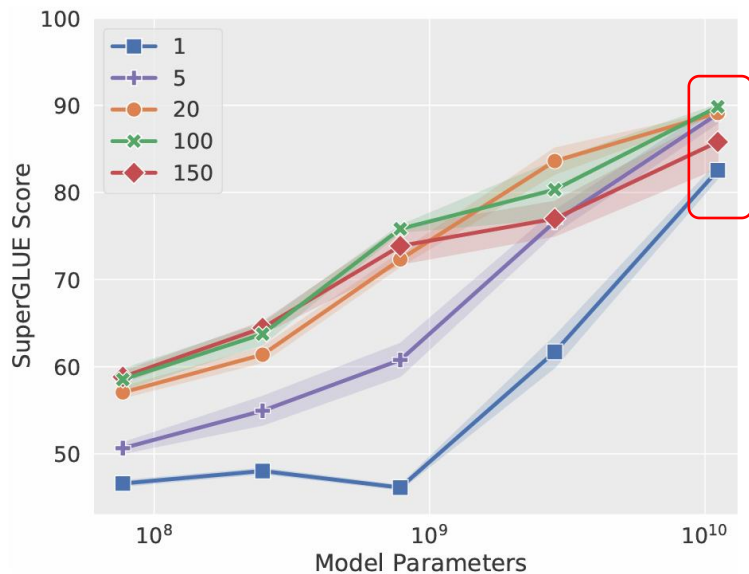


		Virtual tokens		Input embedding vectors			
Mixed task batch	Classify	[.5,...,-7.1]	[2.3, ..., .8]	These	chips	are	tasty
		[.5,...,-7.1]	[2.3, ..., .8]	Elm	Street	is	cool
	Q/A	[1,...,-.4]	[.7, ..., 9.3]	Where	should	I	visit?
	Translate	[.2,...,5.8]	[1.3, ..., -.6]	This	movie	is	boring

可以和全量调参媲美，当参数量>11B
More applicable to larger models



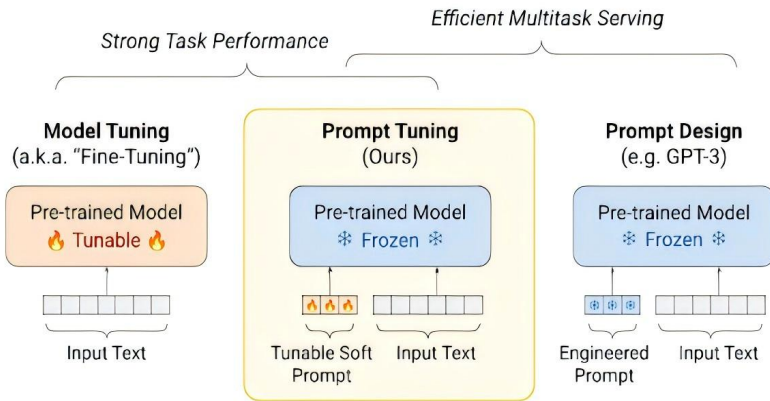
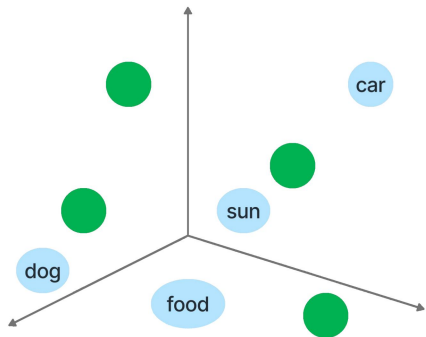
提示词长度对大模型影响更小
Prompt length of 20-100 is typical



总结 优缺点

1. 对于一个新模型自动学习一个新 prompt (反向传播帮助我们找到最佳表达)
2. 只需要一个基础模型
3. 弹性的 domain shift

Find which tokens are nearest to the virtual tokens



可解释性差:

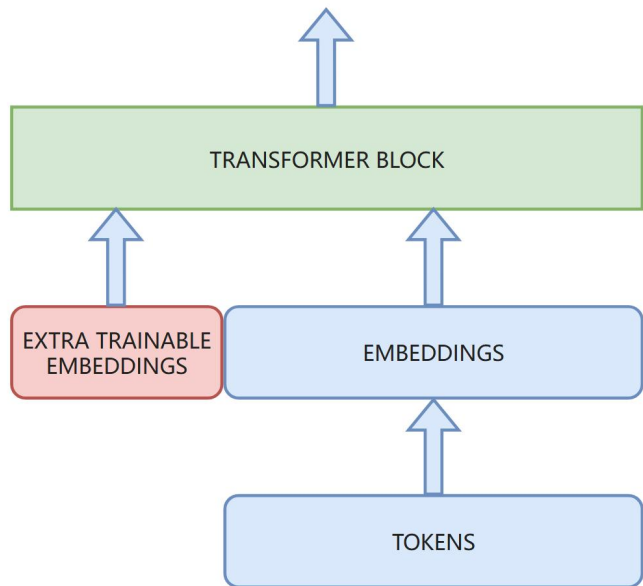
1. 需要转化 embeddings 为 tokens
2. 使用 cosine 距离去找到 top-K nearest neighbors

PEFT 参数高效微调

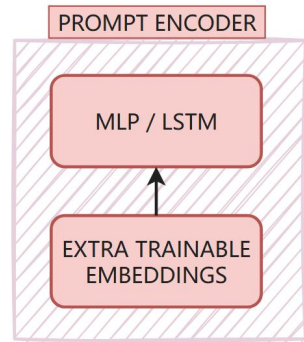
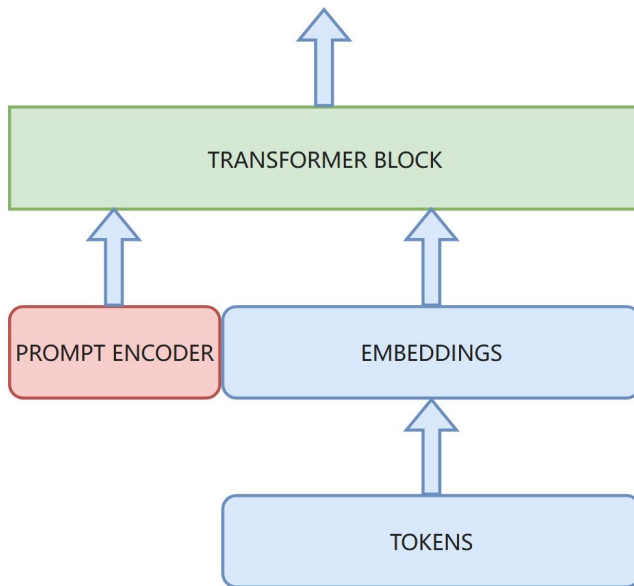


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Prompt Tuning



P Tuning

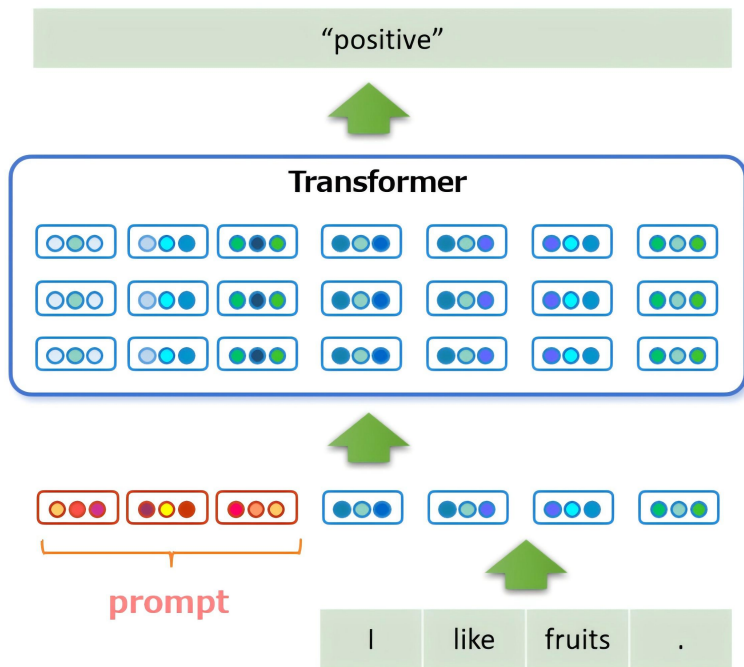


PEFT 参数高效微调

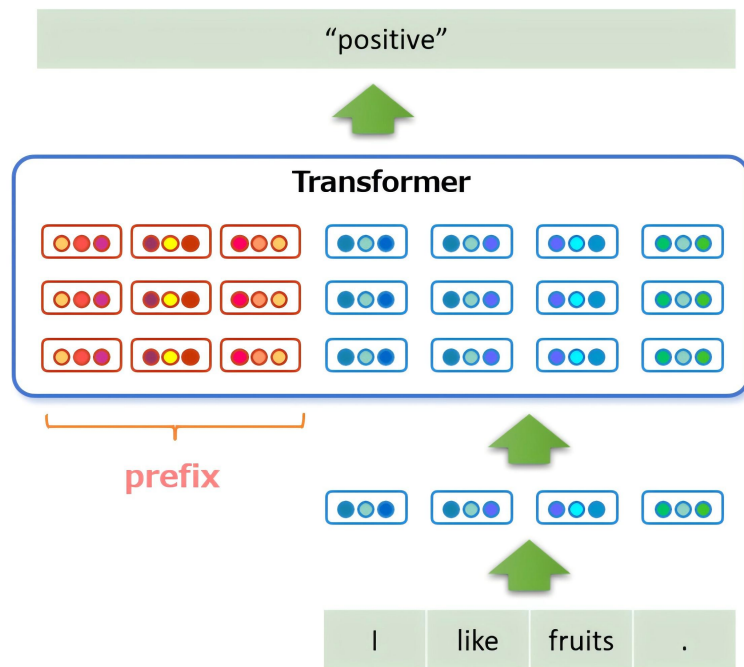


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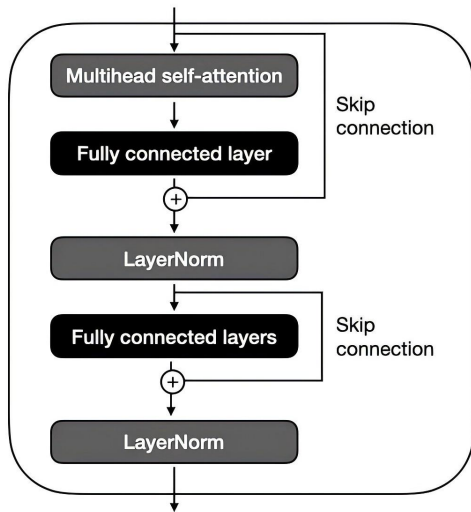
Prompt Tuning



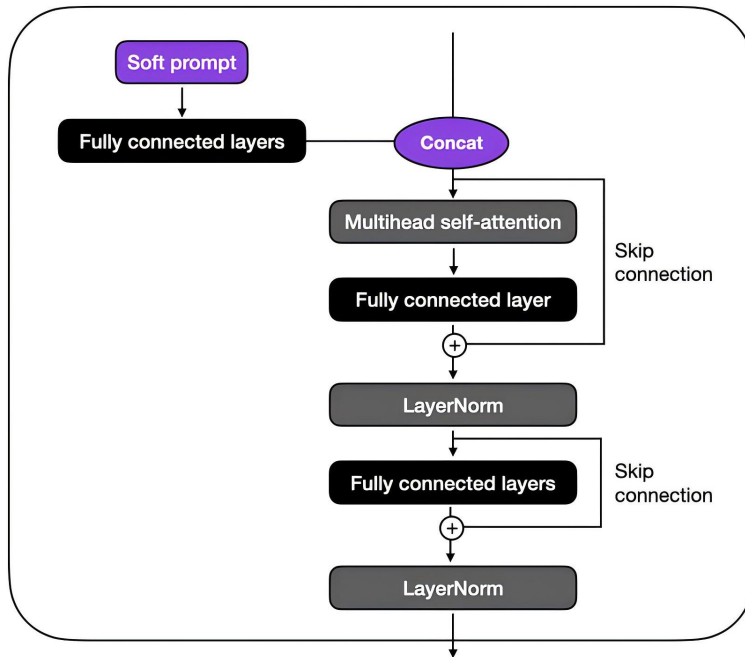
Prefix Tuning

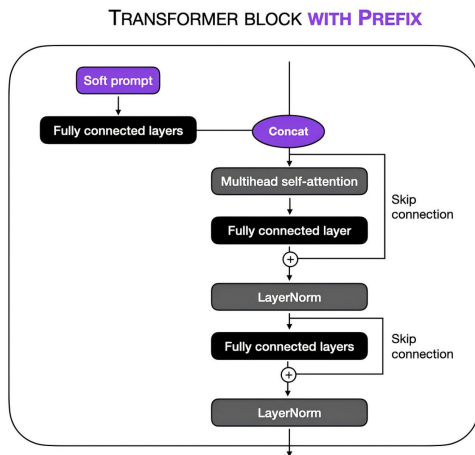


REGULAR TRANSFORMER BLOCK

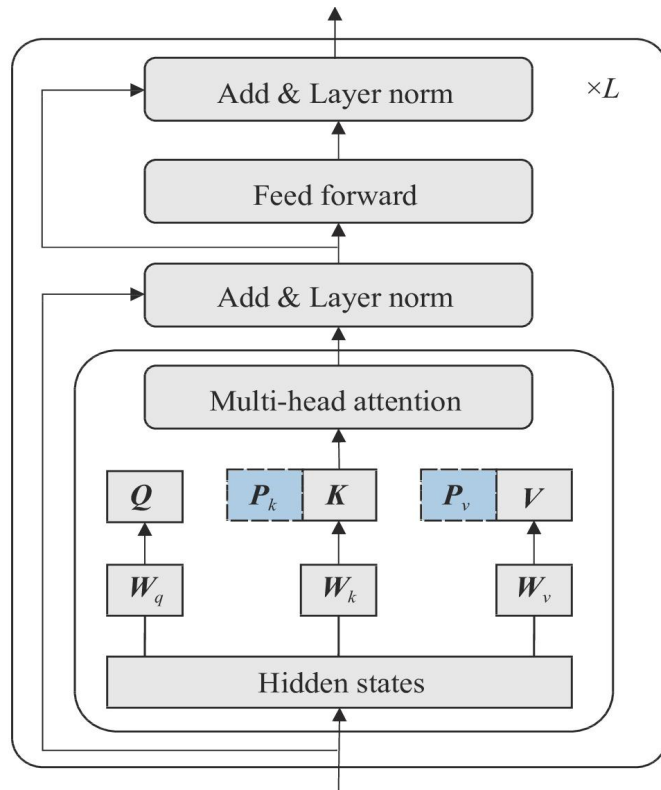


TRANSFORMER BLOCK WITH PREFIX





$$Attention(Q, K, V) = softmax(\frac{Q[P_k, K]^T}{\sqrt{d}})[P_v, V]$$

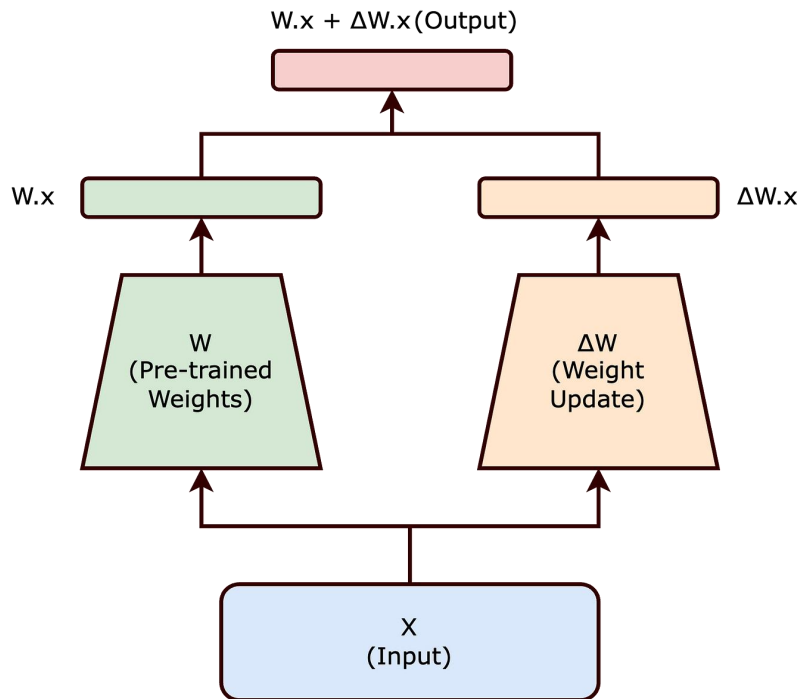
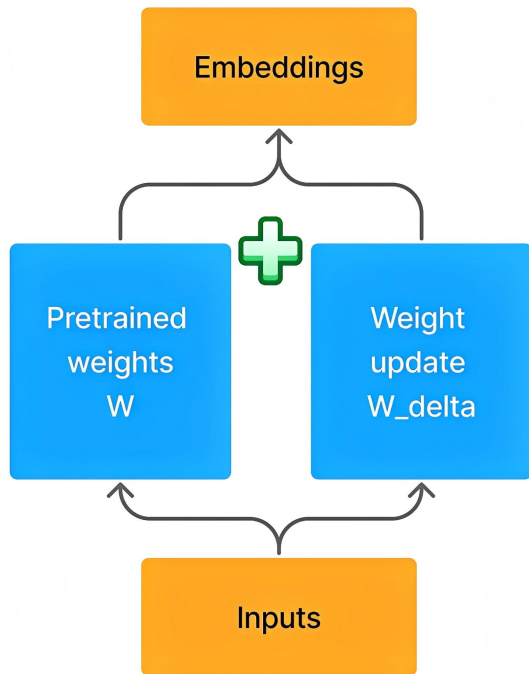


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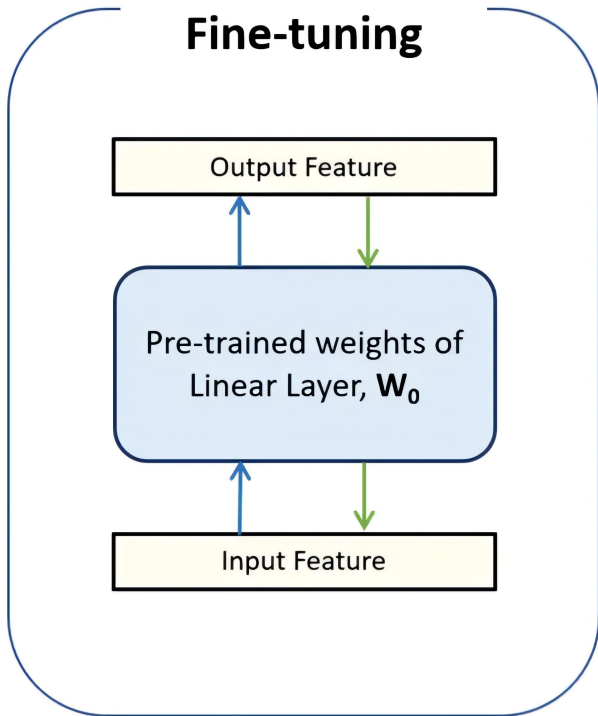


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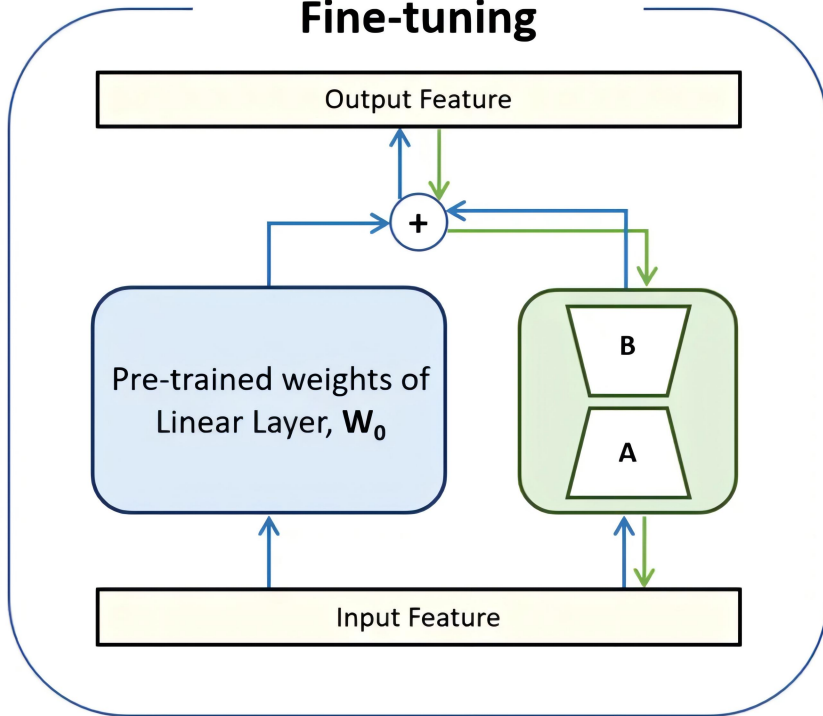
Regular fine-tuning



Full-Parameter Fine-tuning



LoRA Fine-tuning

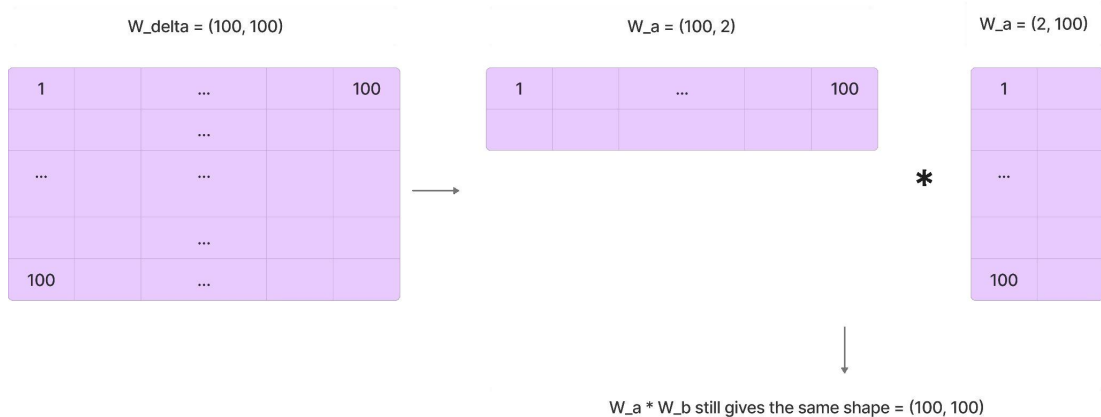
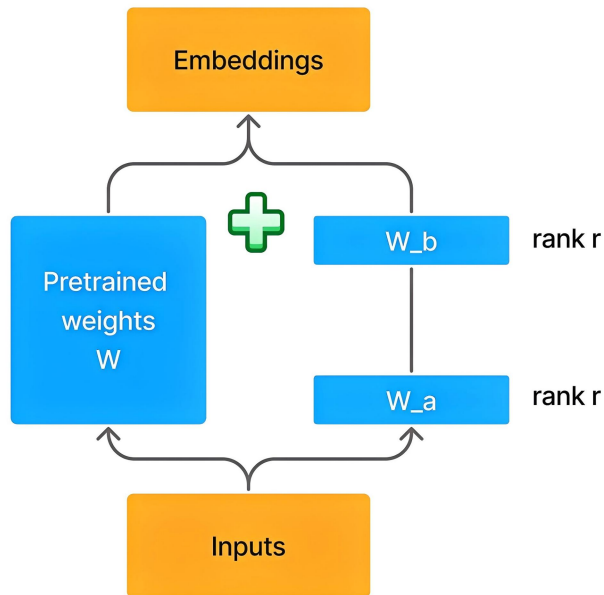


→ forward

→ backward

LoRA

where W_a and $W_b = W_{\Delta}$



Total parameters = $(100 \times 2) + (2 \times 100) = 400$

Original parameters = $(100 \times 100) = 10,000$ parameters

Reduction = $10,000 - 400 = 96\%$!

秩 (线性代数) [\[编辑 \]](#)

条目 [讨论](#) [汉](#) [汉](#) [大陆简体](#) [▼](#)

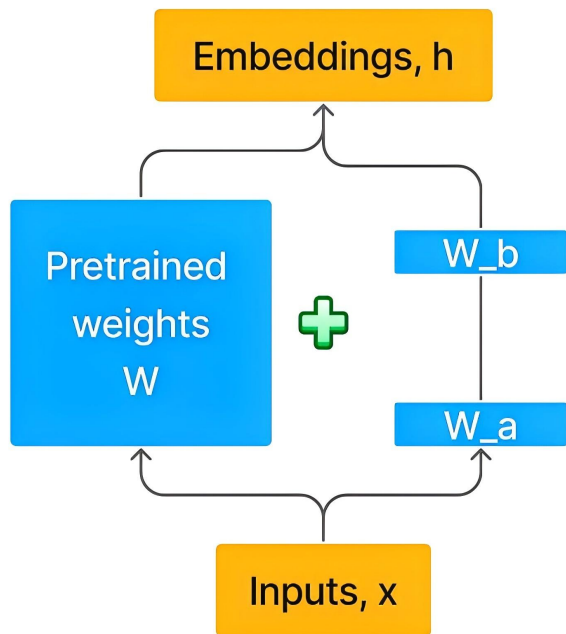
在[线性代数](#)中，一个矩阵 A 的列秩是列向量生成的最大[线性无关](#)组的向量个数。类似地，行秩是矩阵 A 的[线性无关](#)的横行的个数。矩阵的列秩和行秩总是相等的，因此它们可以简单地称作矩阵 A 的**秩** (Rank)。通常表示为 $r(A)$ ， $\text{rank}(A)$ 或 $\text{rk}(A)$ 。

$$\begin{bmatrix} 1 & 2 & 3 \\ 3 & 6 & 9 \end{bmatrix}$$

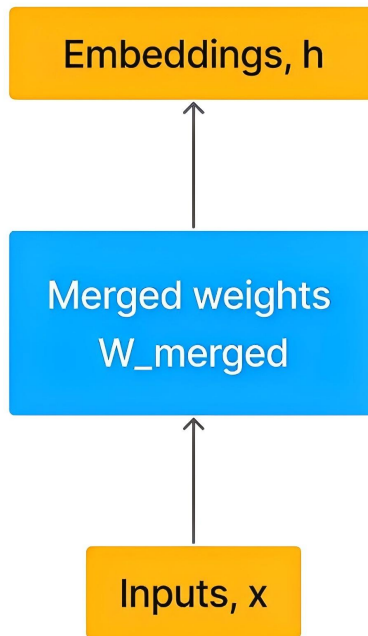
- Row rank: 1
2nd row = 3x 1st row
- Column rank: 1
2nd column = 2x 1st column
3rd column = 2x 2nd column

Observation: Actual rank of the attention weight matrices is low

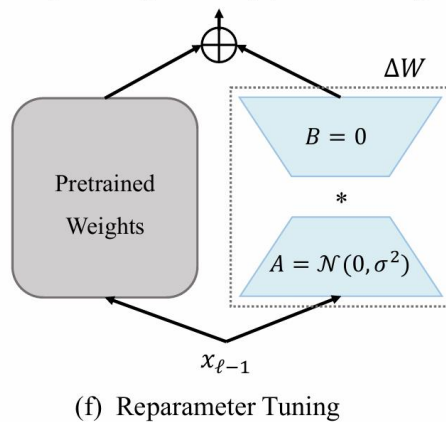
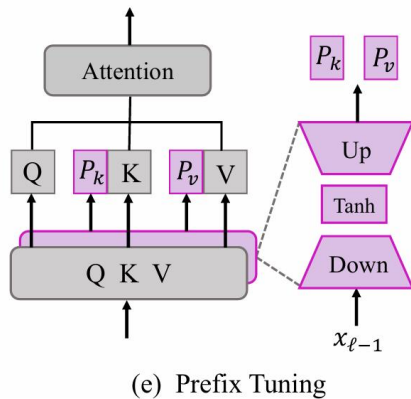
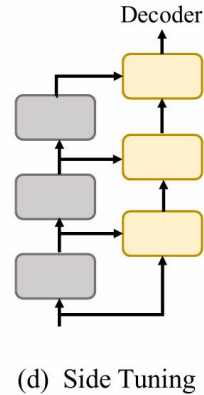
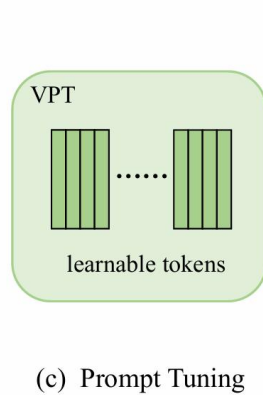
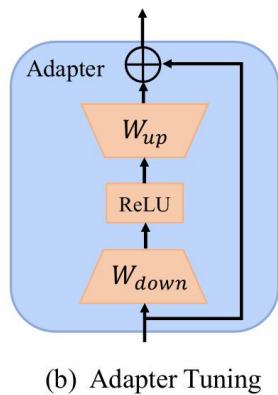
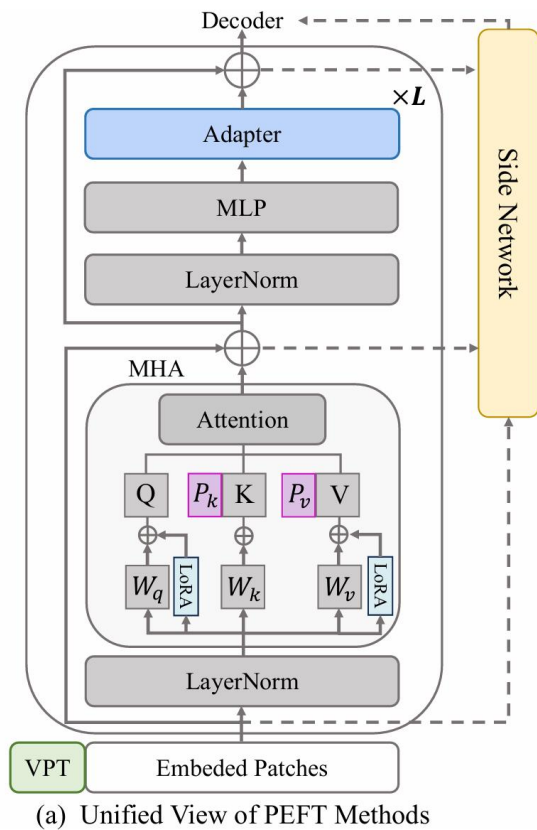
During training



During serving



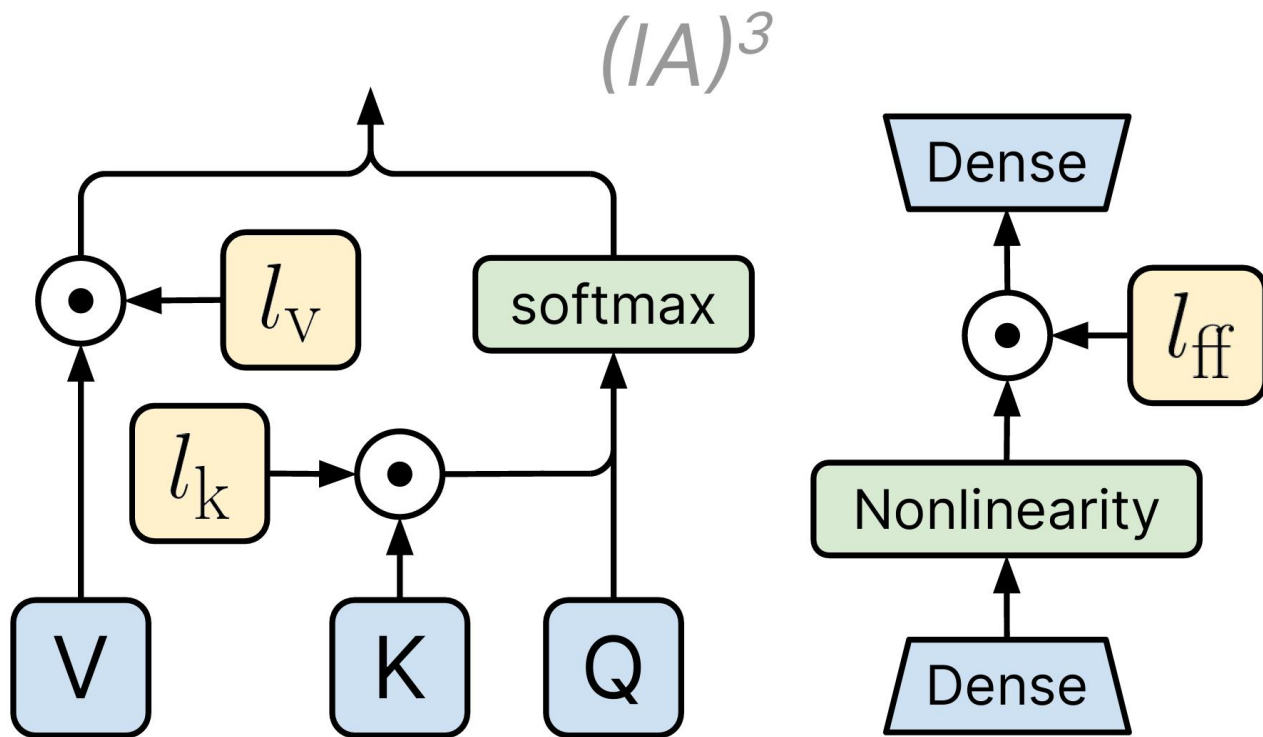
$$h = Wx + (b * a) x$$
$$h = (W + b*a) x$$
$$h = W_merged * x$$



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- PEFT 进阶操作



IA3 - Infused Adapter by Inhibiting and Amplifying Inner Activations

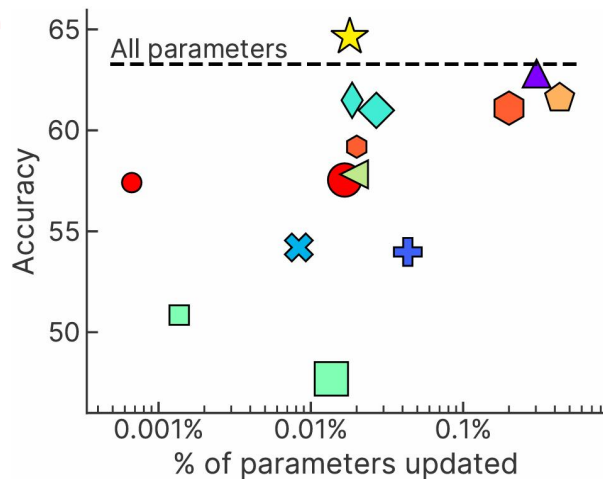


Figure 2: Accuracy of PEFT methods with L_{UL} and L_{LN} when applied to T0-3B. Methods that with variable parameter budgets are represented with larger and smaller markers for more or less parameters.

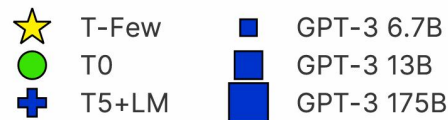
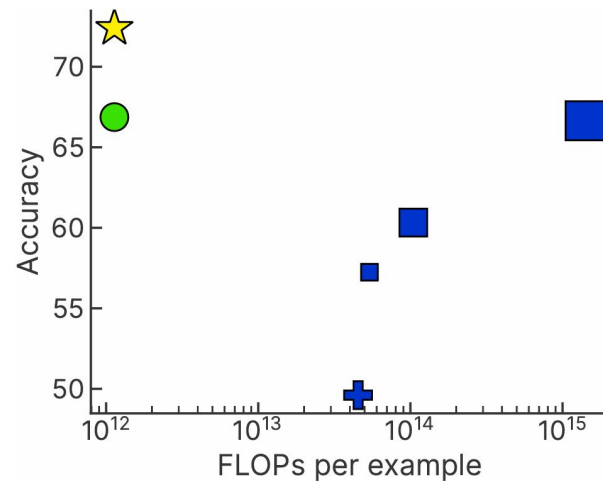


Figure 3: Accuracy of different few-shot learning methods. T-Few uses (IA)³ for PEFT methods of T0, T0 uses zero-shot learning, and T5+LM and the GPT-3 variants use few-shot ICL. The x-axis corresponds to inference costs; details are provided in section 4.2.